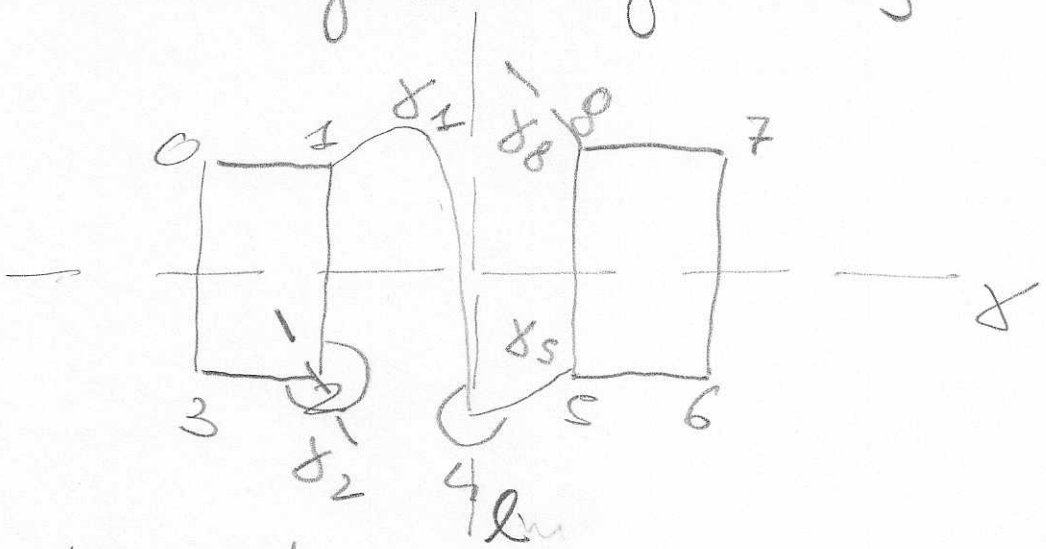
1508

The diagram is given by



We let γ_1 1, γ_2, γ_5 and γ_4 be the equators bisecting 1, 2, 5 and 8

Note that the two pins are antipodal and one is contained in γ_8, while the other one is contained in γ_2,

so $\{P_1, P_2\} \subset \gamma_2 \cap \gamma_8$

Note that the symmetry with respect to γ maps γ_1 in γ_2 and γ_5 in γ_8

So, since γ_1 and γ_5 intersect

transversally in γ (they form angles $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$) we have that

a) $\gamma_2 \cap \gamma_g = \{P_1, P_2\}$

b) γ is mapped by σ in one of the pins, say P_1

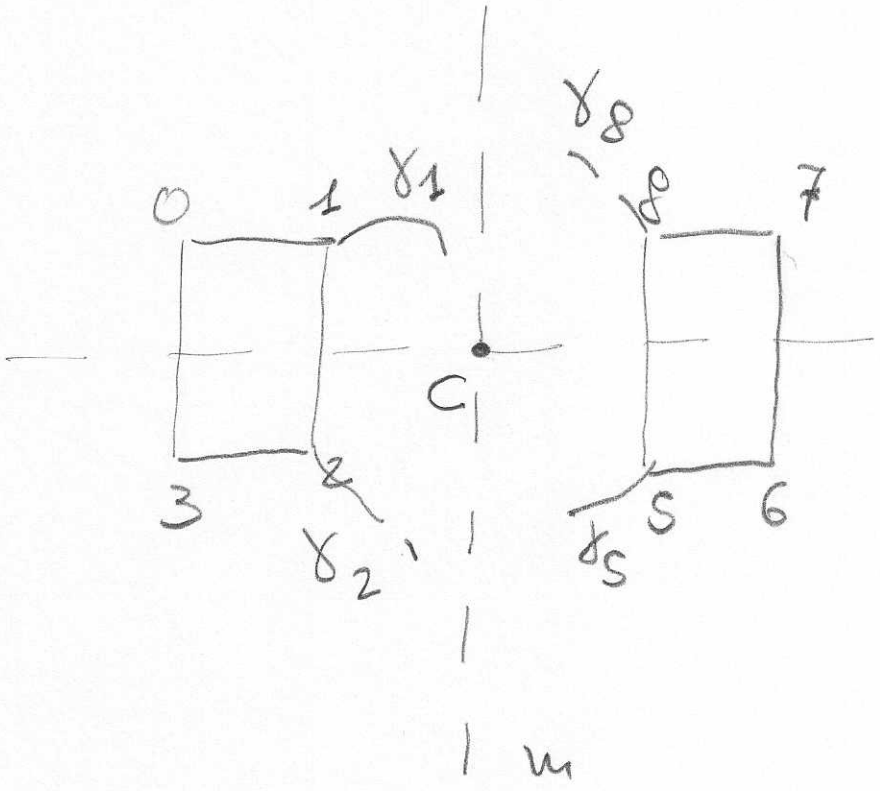
But the bisector ℓ of γ contains both P_1 and P_2. Recall that

c) By the rules to derive the derived-axle net from the original net, γ is different from both P_1 and P_2. So P_1 and γ determine a unique equator going through it, which is precisely ℓ.

By b) ℓ is a meridian

So P_1, P_2 and 4 are on a Hugle meridian.

Consider now the meridian m which



goes through the point $C \in \gamma$ which is equidistant from the two quadrilaterals.

The symmetry with respect to m maps γ_8 into γ_1 and γ_2 into γ_5

So again 4 is mapped by s in one of the P_i's and, because $s^2 = id$, $s(P_i) = 4$. So s maps l into itself and hence $l = m$.

Finally consider the symmetry τ with respect to C. τ maps γ_2 in γ_8 and γ_8 in γ_2. So it maps $\{P_1, P_2\}$ into $\{P_1, P_2\}$

Since these are on the meridian m, either P_1 and P_2 are the north and south pole or they are C and its antipodal.

But recall b): $\sigma(P_1) = 4$, and since σ is the symmetry with respect to γ either

it maps P_1 into itself or P_1 into P_2

So either $q = P_1$ or $q = P_2$, which we already observed in c), it is not possible.